

Q3: Select the correct choice(s) for each of the following points: (Note selecting wrong answers would cause a penalty) (Answer 6) (9 Marks)

1- Describing the digital circuit in hierarchical way is useful because:

- ☒ a. It enable the designer to reuse the some parts of the design
- ☒ b. Divide a complicated circuit into manageable smaller circuit.
- ☐ c. Reduce the size of the implemented circuit.
- ☒ d. Improve performance of the final design

2- For each component in top-level entity:

- ☐ a. It can be declared multiple times
- ☒ b. In can be instantiated multiple times.
- ☒ c. When instantiated, it should have unique label
- ☐ d. When instantiated, mapping its ports is optional.

3- The ASSERT statement is used to check output values against expected values. select the Severity options of this command:

- ☒ a. Note
- ☐ b. Wrong
- ☒ c. Failure
- ☒ d. Error
- ☐ e. Correct
- ☒ f. Warning

4- The initial value that is assigned to Generic in component could be overwritten in the top-level entity

- ☒ a. True
- ☐ b. False

5- Select which statement bellow are legal (more than one choice are allowed)

- ☐ a. $s2(7) \leq s1;$
- ☐ b. $s2 \leq v1 \text{ XOR } "10001000";$
- ☐ c. $s2 \leq v2;$
- ☐ d. $s3 \leq s4(0);$

6- Given the signals bellow, Which of the following is a legal VHDL assignment:

signal X : std_logic_vector(7 downto 0);
signal Y : unsigned(7 downto 0);

- ☒ a. $X \leq \text{std_logic_vector}(Y);$
- ☐ b. $Y \leq \text{std_logic_vector}(X);$
- ☒ c. $X \leq \text{unsigned}(Y);$
- ☐ d. $Y \leq \text{to_integer}(X);$

7- The following VHDL code is for:

```
entity XXX is
port (B, D : in STD_LOGIC;
      Q: out STD_LOGIC);
end XXX;
architecture arch of XXX is
begin
  X: process (B,D)
  begin
    if (B='1') then
      Q <= D;
    end if;
  end process X;
end arch;
```


Describing the digital circuit in hierarchical way is useful because:

Select one or more:

- ☒ a. It enable the designer to reuse the some parts of the design
- ☒ b. Reduce the size of the implemented circuit.
- ☒ c. Divide a complicated circuit into manageable smaller circuit.
- ☐ d. Improve performance of the final design

generics are not synthesized and can be manipulated using non-synthesizable fu

Select the correct statements:

- ☒ a. General-purpose loops are specifically designed for synthesis
- ☒ b. The for-generate loop is specifically designed for synthesis
- ☐ c. General-purpose loops are used in non-synthesizable code such as testbenches
- ☐ d. The for-generate loop are used in non-synthesizable code such as testbenches

to search



SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

SIGNAL s5: INTEGER RANGE -35 TO 35;

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

VARIABLE v2: INTEGER RANGE -35 TO 35;

Select which statement bellow are legal (more than one choice are allowed)

Select one or more:

☐ s2 <= v2;

☒ s3 <= s4(0);

☒ s2 <= v1 XOR "10001000";

☒ s2(7) <= s1;

Initial values for signals in VHDL are:

Select one or more:

☒ a. Is to be avoided in synthesizable code

☐ b. Are to be avoided in Testbench

☐ c. Is to be avoided in non-synthesizable code

☒ d. Are useful in Testbench



Question 4

Not yet answered

Marked out of 1.00

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal
(more than one choice are allowed)

Select one or more:

☒ s2 (5 DOWNT0 0);<= S2 (7 DOWNT0 2);

☐ s2 <= (OTHERS=>'0');

☐ s2 <= v1 XOR '111000';

☒ s3 <= Z;

Which of the following statements
are true regarding testbenches in
* VHDL

Unit under test is declared as
component in the Testbench



synthesizable



Not synthesizable



Does not include input or output
ports



The following VHDL code is for:

entity XXX is

port (B, D : in STD_LOGIC;

Q: out STD_LOGIC);

end XXX;

architecture arch of XXX is

begin

X: process (B,D)

begin

if (B='1') then

Q <= D;

end if;

end process X;

end arch;

Select one:

- ☐ a. D flip Flop with Synchronous reset
- ☐ b. D flip Flop with Synchronous reset and enable
- ☐ c. D flip Flop with Asynchronous clear
- ☐ d. D latch with enable
- ☐ e. D flip Flop with Asynchronous clear and enable

Question **5**

Not yet answered

Marked out of 1.00

🚩 Flag question

To design a 9 to 0 counter we will need how many D-flip flops?

Select one:

- ☐ a. 3
- ☐ b. 4
- ☐ c. 6
- ☐ d. 5

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Next page



Question 4

Not yet answered

Marked out of 1.00

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal
(more than one choice are allowed)

Select one or more:

- ☐ s2 (5 DOWNT0 0);<= S2 (7 DOWNT0 2);;
- ☐ s2 <= (OTHERS=>'0');
- ☐ s2 <= v1 XOR "111000";
- ☐ s3 <= 'Z';



Question 2

Not yet answered

Marked out of 1.00

Flag question

Given the signals bellow

signal X : std_logic_vector(7 downto 0);

signal Y : unsigned(7 downto 0);

Which of the following is a legal VHDL assignment

Select one:

- ☐ a. Y <= to_unsigned(201, 8);
- ☐ b. X <= unsigned(Y);
- ☐ c. Y <= std_logic_vector(X);
- ☐ d. X <= to_integer(Y);



العربية

الفرنسية





Question 1

Not yet answered

Marked out of 1.00

Flag question

Say that $a(7:0)$ "00110011" and $b(3:0) = "1111"$.
Determine the values produced by the assignments below:

$a(7 \text{ DOWNTO } 4) \text{ NAND } "0111"$

Answer:

Next page



العربية

الفرنسية

Question 2

Incorrect

Mark 0.00 out of 1.00

🚩 Flag question

Say that $a(7:0)$ "00110011" and $b(3:0) =$ "1101".
Determine the values produced by the assignments below:

$a(7 \text{ DOWNTO } 4) \text{ XOR NOT } b;$

Answer: 0101





Question 25

Answer saved

Marked out of 0.60

Flag question

To design a counter that counts the even numbers from 0 to 20, how many D-flip flops we will need ?

Select one:

- ☐ a. 6
- ☐ b. 5
- ☒ c. 4
- ☐ d. 3

[Clear my choice](#)

Previous page

Finish attempt ...





Marked out of 0.50

Flag question

The initial value that is assigned to Generic in component could be overwritten in the top-level entity:

Select one:

☒ True

☐ False

Question 4

Not yet answered

Marked out of 1.50

Flag question

Using the state transition table as a guide, label each transition (T1-T7) with IN/OUT (ex: 0/00) to represent





Question 7

Not yet answered

Marked out of 0.50

Flag question

For the FSM Mealy mode outputs of a circuit are a function of:

Select one:

- ☐ a. the present inputs of the circuit.
- ☐ b. the next state
- ☐ c. the present state.
- ☐ d. the next state and the present inputs of the circuit.
- ☒ e. the present state and the present inputs of the circuit.

Question 8

Not yet answered





In the following VHDL architecture code :

with SEL select

z <= A when "00",

B when "01",

C when "10",

D when others;

The when others covers and catches the following conditions:

Select one:

☐ a. when "101"

☒ b. when "11"

☐ c. when "00"

☐ d. when '1'

Question 14

Not yet answered



10





Flag question

Determine which lines have syntax errors in the accompanying VHDL code:

(Note selecting wrong answers would cause a penalty)

```
1 library ieee;
2 use ieee.std_logic_1164.all;
3 entity Error_finder is
4     generic
5         (NUM_STAGES : natural := 16);
6     port
7         ( clk, enable, reset, sr_in: std_logic;
8           sr_out : out std_logic);
9 end entity;
10 architecture rtl of Error_finder is
11     type sr_length is array ((NUM_STAGES-1) to 0) of std_logic;
12     signal sr: sr_length;
13 begin
14     process (clk, reset)
15     begin
16         if (reset = '1') then
17             sr <= (others=>'0');
18         elsif (rising_edge(clk)) then
19             if (enable = '1') then
20                 sr((NUM_STAGES-1) downto 1) <= sr((NUM_STAGES-2) downto 0);
21                 sr(0) <= sr_in;
22             end if;
23         end process;
24         sr_out <= sr(NUM_STAGES-1 downto 0);
25     end rtl;
```

Select one or more:

- ☒ a. line 8
- ☐ b. Line 2
- ☐ c. Line 18
- ☒ d. Line 9



10





Question 22

Not yet answered

Marked out of 0.50

Flag question

In the FPGA design flow, Timing based simulation occurs before synthesis

Select one:

☐ True☒ False

10



Consider the Shift Register with Load shown in the figure below. When load = '0', it operates as a circular shift register. However, if load = '1', data is loaded into the DFFs at the next positive clock transition (thus the initial state of the flip-flops can be programmed). Assuming you have the





Marked out of 0.60

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal (more than one choice are allowed)

Select one or more:

☐ s1 <= "11";

☒ s2(3 to 0) <= s4(5 to 2);

☒ s2 (6 DOWNT0 1) <= S2 (7 DOWNT0 2);

☐ s4(4 downto 3) <= s3 & s2(1);





Does not include input or output ports



إرسال

Given the signals bellow

signal X : std_logic_vector(7 downto 0);

signal Y : integer;

Which of the following is a legal VHDL assignment

Select one:

- ☐ a. Y <= to_integer(unsigned(X));
- ☐ b. X <= to_integer(Y);
- ☒ c. Y <= to_integer(X);
- ☐ d. X <= to_integer(unsigned(Y));

Consider the full adder shown in the figure bellow. Select the right choices from the drop-list of the code top-level design in VHDL in order to have **8-bit full adder**





Name the parts of the structure of a a VHDL file?

Select one:

- ☐ a. Entity, and architecture
- ☐ b. Library, Entity, Process, Components, and Architecture
- ☒ c. Library, Entity, and architecture
- ☐ d. Entity
- ☐ e. Library, Entity, process, and architecture

Clear my choice

* ?Why is hierarchical design useful

Segment a complex design into smaller pieces



Ability to reuse common elements





Clear my choice

58 من 65

* ?Why is hierarchical design useful

Segment a complex design into smaller pieces



Ability to reuse common elements



More maintainable design



All the above



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تم إنشاء هذا النموذج داخل University of Kufa. الإبلاغ عن إساءة الاستخدام





☐ d. It can be declared multiple

65 من 65

The V in VHDL stands for?

Select one:

☐ a. Version 5

☒ b. Very

☐ c. Vast

☐ d. Very High Speed IC

☐ e. Verilog

n :)





تم اسئله معماريه عمار وسلام....

d. 6

The ASSERT statement is used to check output values against expected command:

Select one or more:

- ☒ a. Note
- ☒ b. failure
- ☐ c. Wrong
- ☒ d. warning
- ☒ e. Error
- ☐ f. Correct

ge

e here to search





- ☐ d. It can be declared multiple times

The following VHDL statements deduce a flip-flop :

Select one or more:

- ☒ a. if (clk'event and clk='1') then q<= d;
- ☒ b. if (rising_edge(clk)) then q<= d;
- ☐ c. if (clk= '1') then q<= d;
- ☐ d. q<= d when (clk='1') else q<= 0;

ere to search



C when "10".

D when others:

The when others covers and catches the following conditions





Photo

بنونه *-*

end XXX;

architecture arch of XXX is

begin

X: process (B, R)

begin

if (R='1') then

Q <= '0';

else

if (rising_edge(B)) then

Q <= N;

end if;

end if;

end process X;

end arch;

Select one:

- ☐ a. 4-bit register without clear, reset and enable
- ☐ b. 4-bit register with Synchronous reset and enable
- ☐ c. 4-bit register with Synchronous reset
- ☒ d. 4-bit register with Asynchronous clear





Question 24

Not yet answered

Marked out of 0.60

Flag question

Given the signals bellow

signal X : std_logic_vector(7 downto 0);

signal Y : unsigned(7 downto 0);

signal Z : integer;

Which of the following is a legal VHDL assignment

Select one or more:

- ☐ a. X <= to_integer(Y);
- ☐ b. X <= unsigned(Y);
- ☒ c. Y <= to_unsigned(35, 4);
- ☐ d. Y (5 downto 0) <= to_unsigned(17, 6);
- ☒ e. Z <= to_integer(unsigned(x));

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Finish attempt

Question 7

Not yet answered

Marked out of 1.00

Flag question

Say that a(7 downto 0) = "11110011" and b(3 downto 0) = "1001". Determine the values produced by the assignments below:

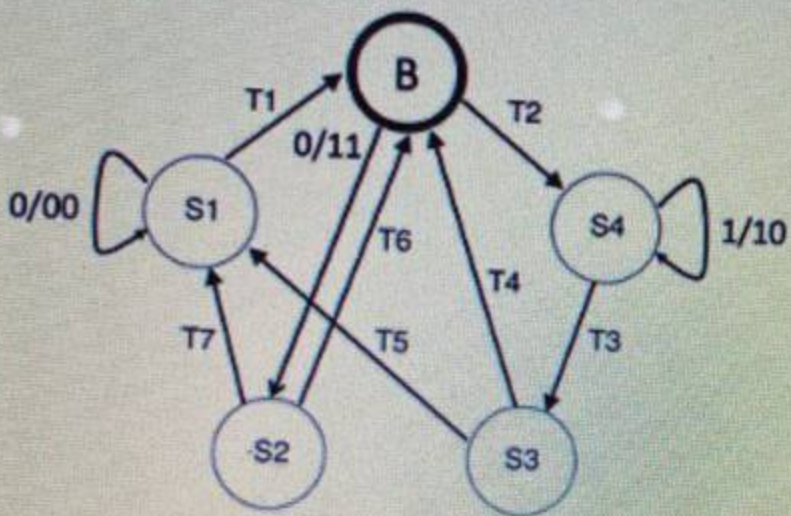
b(5 DOWNTO 2) <= a(7 DOWNTO 4);

Answer: 0101





E	0	A	10
E	1	B	01



T7	0/01	⇕
T2	1/00	⇕
T3	0/01	⇕
T6	1/01	⇕
T1	1/11	⇕
T5	0/10	⇕
T4	1/01	⇕

